

LOUIS NYIRONGO

SENIOR SOFTWARE ENGINEER | DIGITAL HEALTH | INTEROPERABILITY | DEVOPS

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PROFESSIONAL PROFILE

Senior Software Engineer and DevOps practitioner with eight years of experience spanning digital health, government systems, cloud infrastructure, Linux administration and full-stack product delivery. His work since 2020 has focused on strengthening Malawi's national health information systems, including MaHIS/OpenMRS, DHIS2, OpenHIM, OpenLMIS, facility registries, terminology services and public-health dashboards. Combines hands-on engineering with stakeholder engagement, interoperability design, production deployment, technical documentation and partner support.

CORE EXPERTISE

Digital health and interoperability: MaHIS/OpenMRS, DHIS2, OpenHIM, OpenLMIS, OpenIMIS, health facility registries, terminology services, EMR workflows, aggregate data exchange

Software engineering: JavaScript, TypeScript, Python, C#, React, Ionic, Next.js, Angular, Vue, Node.js, NestJS, Django REST Framework, Ruby on Rails, Laravel, ASP.NET MVC

Cloud, DevOps and data: AWS, Google Cloud Platform, Docker, Kubernetes, Ansible, CircleCI, Linux, Nginx, shell scripting, MySQL/MariaDB, PostgreSQL, SQL Server, MongoDB, REST APIs, Jira

PROFESSIONAL EXPERIENCE

Luke International Norway | Lilongwe, Malawi

Senior Software Engineer - AHD Improvement through MaHIS Strengthening (AIMS) | 2024-Present | Project period: 2024-2027

Bill & Melinda Gates Foundation-funded programme strengthening advanced HIV disease screening, care and treatment through MaHIS.

- Enhance MaHIS EMR core modules for advanced HIV disease screening, care and treatment, while supporting continuity of HIV-related data across inpatient, outpatient and emergency service points.
- Develop and maintain Vue and Ruby on Rails/OpenMRS workflows, translating clinical and programme requirements into usable modules, validations and service integrations.
- Work with Ministry of Health teams and implementation partners to troubleshoot production issues, refine workflows, document solutions and strengthen system reliability.
- Support interoperability and consistent data exchange across service-delivery points beyond the HIV clinic.

Luke International Norway | Lilongwe, Malawi

Digital Health Technical Support Officer - COSMO Project | 2021-2023

Core Support to Malawi Ministry of Health Digital Health Division; funded by the Bill & Melinda Gates Foundation.

- Provided embedded technical capacity to Ministry of Health personnel responsible for national digital health platforms and services.
- Patched, enhanced and upgraded the Malawi Health Facility Registry, National COVID-19 Dashboard, DHIS2 custom applications and integrations between national systems and partner applications.
- Supported production deployments across cloud and Ministry data-centre infrastructure, including Linux/Nginx configuration, issue resolution and operational documentation.
- Collaborated with government teams and partners including GIZ, VillageReach, World Vision and UNICEF on implementation and technical support needs.

Luke International Norway | Lilongwe, Malawi

Software Developer - Kuunika Project | 2020-2022

A four-year Government of Malawi programme funded by the Bill & Melinda Gates Foundation to improve routine data quality and data use.

- Developed and maintained national digital health products, including the COVID-19 Dashboard, Health Facility Registry, National Terminology Service and Malawi COVID-19 vaccine digital certification platform.
- Delivered full-stack features, APIs, data workflows and production fixes in collaboration with Ministry technicians and programme partners.
- Contributed to systems that made routinely available health data more accessible to technical teams and policy makers.

Malawi Revenue Authority | Malawi

Full-Stack / Systems Developer | Nov 2019-Feb 2020

- Developed and maintained tax and revenue-related web applications using C#, ASP.NET MVC, SQL Server, HTML/CSS and JavaScript.
- Implemented server-side logic and front-end workflows with attention to security, usability, efficiency and organisational compliance requirements.
- Collaborated with cross-functional teams to translate operational needs into maintainable system features.

MIHR Africa | Lilongwe, Malawi

Software Developer | Jan 2019-Oct 2019

- Built an integrated Human Resource Management System tailored to client requirements using C#, ASP.NET MVC, SQL Server, HTML/CSS and JavaScript.
- Developed front-end and back-end functionality for employee management, payroll and performance-management processes.
- Contributed to system design, implementation and optimisation to improve functionality and user experience.

Malawi Sustainable Development Network Programme (SDNP) | Blantyre, Malawi

Linux Systems Administrator Intern | Feb 2018-Dec 2018

- Administered the FRED client database supporting .MW domain registrations and renewals.
- Monitored DNS, email and web-hosting services to support availability and reliability.
- Automated routine administration using shell scripts and optimised billing processes to reduce execution time.
- Resolved network and server issues for internal and external clients.

SELECTED CONSULTANCY AND INTEGRATION PROJECTS

TotalEnergies Malawi Workflow System

Backend engineering and cloud migration consultancy

- Enhanced a Django REST Framework backend and migrated the workflow platform from Hostinger to Google Cloud Platform, including file-storage reconfiguration, Docker-based deployment, API endpoint changes and additional backend features.

Ministry of Health Official eLearning Platform

Moodle enhancement consultancy funded by CHAI

- Developed a hierarchy plugin and modernised the Moodle user interface to improve platform structure and the online learning experience.

Health Services Joint Fund Pricing Registry

OpenIMIS customisation for Christian Health Association of Malawi facilities

- Customised OpenIMIS and developed a dedicated administration portal to automate service capture, pricing and invoice preparation for more than 195 CHAM facilities serving areas without government facilities.

Global Fund Aggregate Data Exchange (ADEX)

DHIS2/OpenHIM interoperability mediator

- Designed, developed and deployed a NestJS mediator that transforms aggregate data from Malawi's national health information system and securely transmits it to the Global Fund data repository through the national interoperability layer.
- Integrated the mediator with a customised DHIS2 application and configured it for reliable handling of large datasets and production operations.

Global Fund - Ministry of Health Interoperability Layer Enhancement

OpenLMIS product catalogue and master-facility integration

- Integrated the national product catalogue with the Global Fund repository through Malawi's health interoperability layer, using the Master Health Facility Registry as the authoritative facility source.
- Supported commodity and procurement management, inventory and warehouse operations, order fulfilment and financial-reconciliation data flows.

National COVID-19 WhatsApp Chatbot

Ministry of Health and VillageReach collaboration

- Designed Botpress conversation flows and Node.js processing scripts to collect and disseminate COVID-19 information, transmit user-submitted data to the One Health Surveillance Platform (DHIS2) and support analytics dashboards.

EDUCATION

Malawi University of Business and Applied Sciences (MUBAS) | Blantyre, Malawi

Bachelor of Science in Management Information Systems | Apr 2014-Dec 2018

PROFESSIONAL DEVELOPMENT

- OpenMRS Conference 2024 - Certificate of Attendance
- Cloud DevOps Engineer - Udacity, Nov 2022-Feb 2023
- ALX-T Cloud Developer - Udacity, Jun-Dec 2022
- DHIS2 Fundamentals - DHIS2 Online Academy, Jun-Dec 2022
- Kubernetes for Beginners - Hands-On - Udemy

ADDITIONAL INFORMATION

Nationality: Malawian

Languages: English and Chichewa